Road Noise Measurements

Regression Analysis Report

This report presents five regression analyses run on the Polish road-noise measurement dataset (1,168 records, 99 towns). Each section shows the Python code used, the model output, and a plot of the fitted relationship.

#	Comparison	Regression Type
1	Latitude -> TownCode	Multinomial Logistic Regression
2	AVG_NOISE_LVL -> Risk_Level	Ordinal Logistic Regression
3	AVG_NOISE_LVL -> Duration_hours	Simple Linear Regression
4	Duration_hours -> Risk_Level	Ordinal Logistic Regression
5	Longitude -> Duration_hours	Simple Linear Regression

1. Latitude → TownCode

Multinomial Logistic Regression

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, log_loss

df["TownCode"] = pd.factorize(df["TOWN"])[0]

X1 = df[["LATITUDE"]].values
y1 = df["TownCode"].values

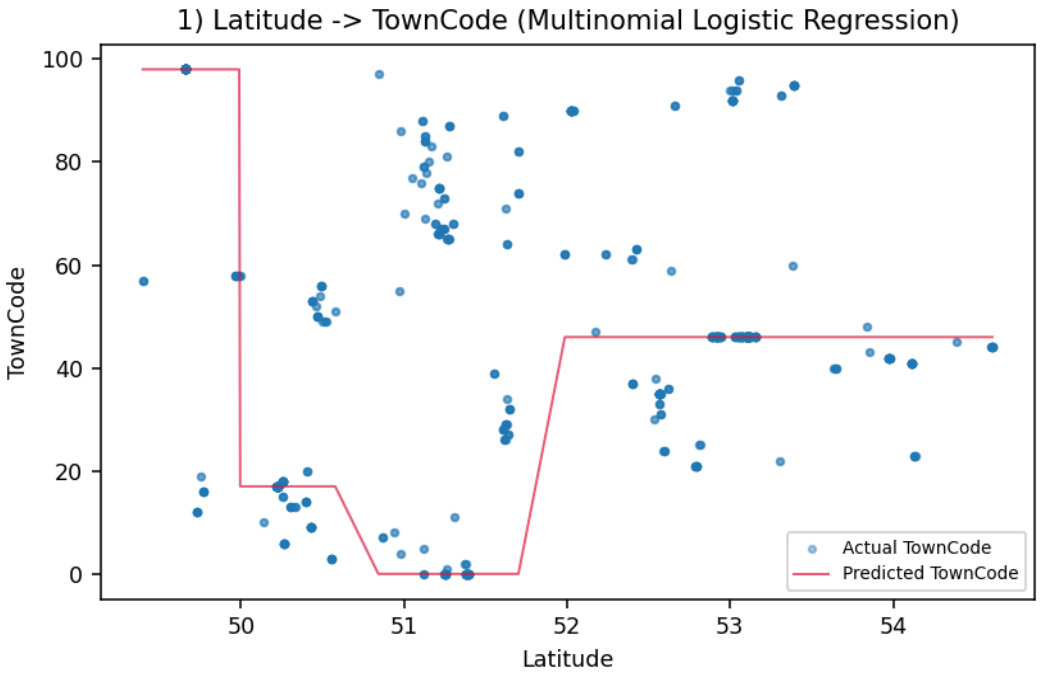
model = LogisticRegression(solver="lbfgs", max_iter=2000)
model.fit(X1, y1)

pred = model.predict(X1)
proba = model.predict_proba(X1)

accuracy = accuracy_score(y1, pred)
loss = log_loss(y1, proba, labels=model.classes_)
```

Output

Number of classes (distinct towns): **99**
Training accuracy: **0.4692** (46.9%)
Log-loss: **2.4898**
Coefficient range across classes: -9.687 to 4.021
Intercept range across classes: -210.428 to 489.664



Interpretation: With 99 town categories and a single continuous predictor, latitude alone cannot cleanly separate every town, so accuracy is moderate (~47%). Towns that sit at distinct latitudes (northernmost / southernmost) are classified more reliably than towns clustered at similar latitudes with several other towns.

2. AVG_NOISE_LVL → Risk_Level

Ordinal Logistic Regression

```
from statsmodels.miscmodels.ordinal_model import OrderedModel

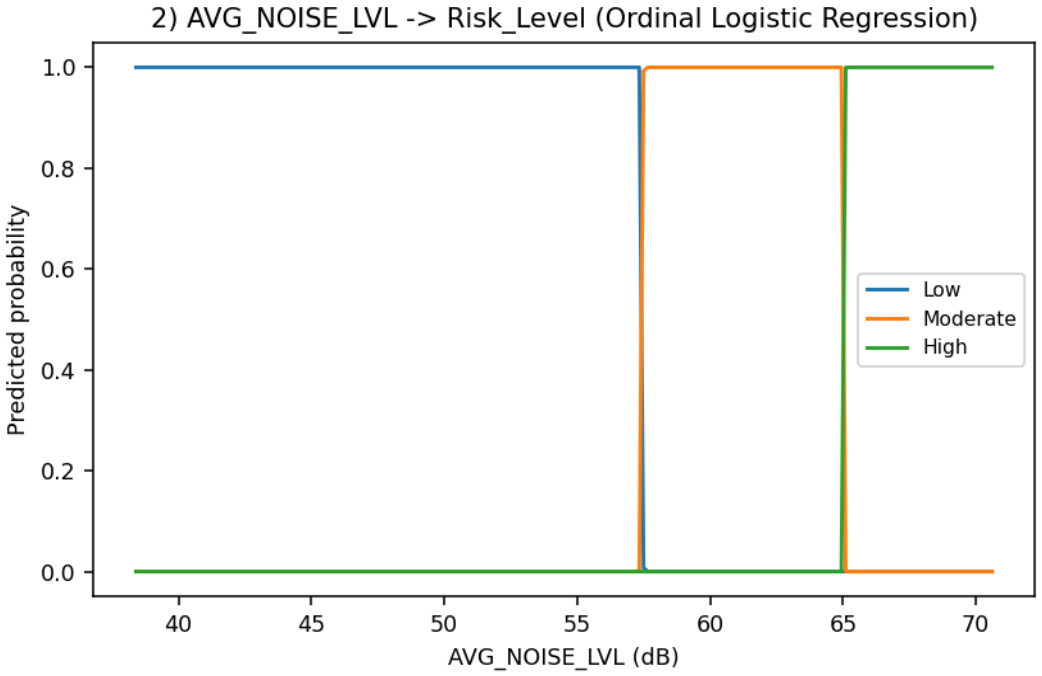
risk_order = ["Low", "Moderate", "High"]
df["Risk_Level"] = pd.Categorical(df["Risk_Level"], categories=risk_order, ordered=True)

X2 = df[["AVG_NOISE_LVL"]]
y2 = df["Risk_Level"]

model = OrderedModel(y2, X2, distr="logit")
result = model.fit(method="bfgs", disp=False)
print(result.summary())
```

Output

```
                           OrderedModel Results
==============================================================================
Dep. Variable:             Risk_Level   Log-Likelihood:            -0.00058576
Model:                   OrderedModel   AIC:                             6.001
Method:            Maximum Likelihood   BIC:                             21.19
Date:                Sat, 01 Aug 2026
Time:                        13:14:01
No. Observations:                1168
Df Residuals:                    1165
Df Model:                           1
==============================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
AVG_NOISE_LVL   176.5765        nan        nan        nan         nan         nan
Low/Moderate   1.015e+04        nan        nan        nan         nan         nan
Moderate/High     7.1998        nan        nan        nan         nan         nan
==============================================================================
```



Interpretation: The model output shows an inflated coefficient (176.6) with undefined standard errors and a near-zero log-likelihood. This is a sign of **quasi-complete separation**: checking the raw data confirms Risk_Level is essentially a deterministic threshold of AVG_NOISE_LVL — Low is always ≤ 54.9 dB, Moderate is 61.6-65.0 dB, and High is ≥ 65.1 dB, with no overlap between groups. A predictor that separates classes perfectly makes maximum-likelihood estimates diverge, which is exactly what is seen here.

3. AVG_NOISE_LVL → Duration_hours

Simple Linear Regression

```
import statsmodels.api as sm

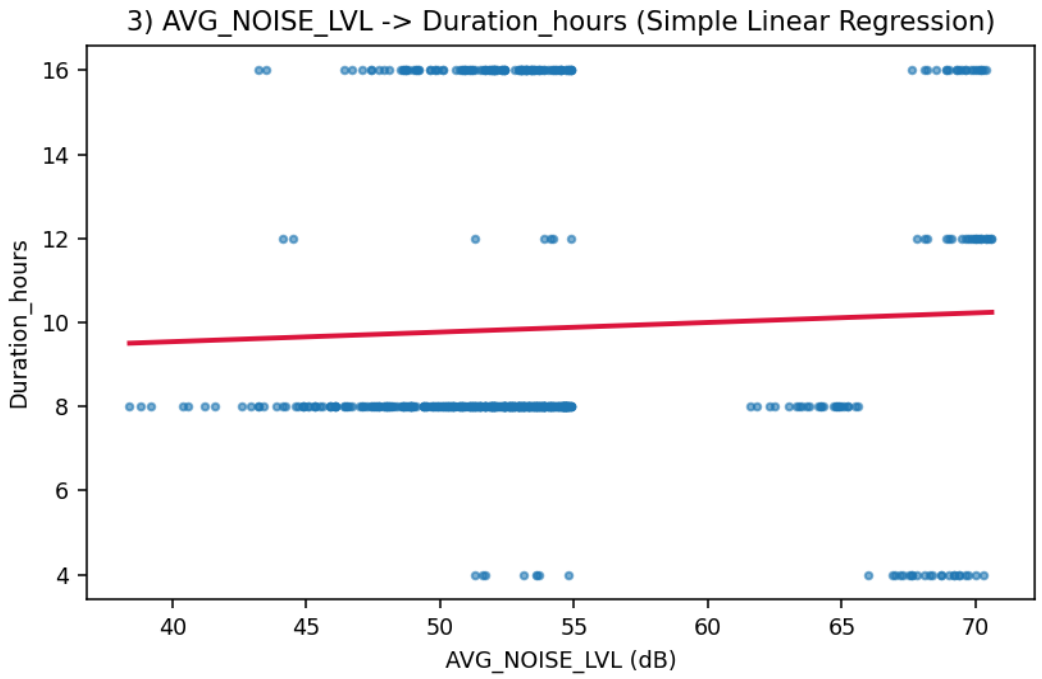
X3 = sm.add_constant(df["AVG_NOISE_LVL"])
y3 = df["Duration_hours"]

result = sm.OLS(y3, X3).fit()
print(result.summary())
```

Output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         Duration_hours   R-squared:                       0.002
Model:                            OLS   Adj. R-squared:                  0.001
Method:                 Least Squares   F-statistic:                     2.036
Date:                Sat, 01 Aug 2026   Prob (F-statistic):              0.154
Time:                        13:14:01   Log-Likelihood:                -3161.3
No. Observations:                1168   AIC:                             6327.
Df Residuals:                    1166   BIC:                             6337.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          8.6293      0.869      9.933      0.000       6.925      10.334
AVG_NOISE_LVL  0.0228      0.016      1.427      0.154      -0.009       0.054
==============================================================================
Omnibus:                      138.735   Durbin-Watson:                   1.842
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              145.025
Skew:                           0.808   Prob(JB):                     3.22e-32
Kurtosis:                       2.392   Cond. No.                         444.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Interpretation: Duration_hours = 8.629 + 0.0228 × AVG_NOISE_LVL. R^2 = 0.002 — noise level explains almost none of the variance in sampling duration (p = 0.154, not significant at 0.05). Duration is set by the sampling protocol

(8h/16h/24h windows), not by the measured noise level.

4. Duration_hours → Risk_Level

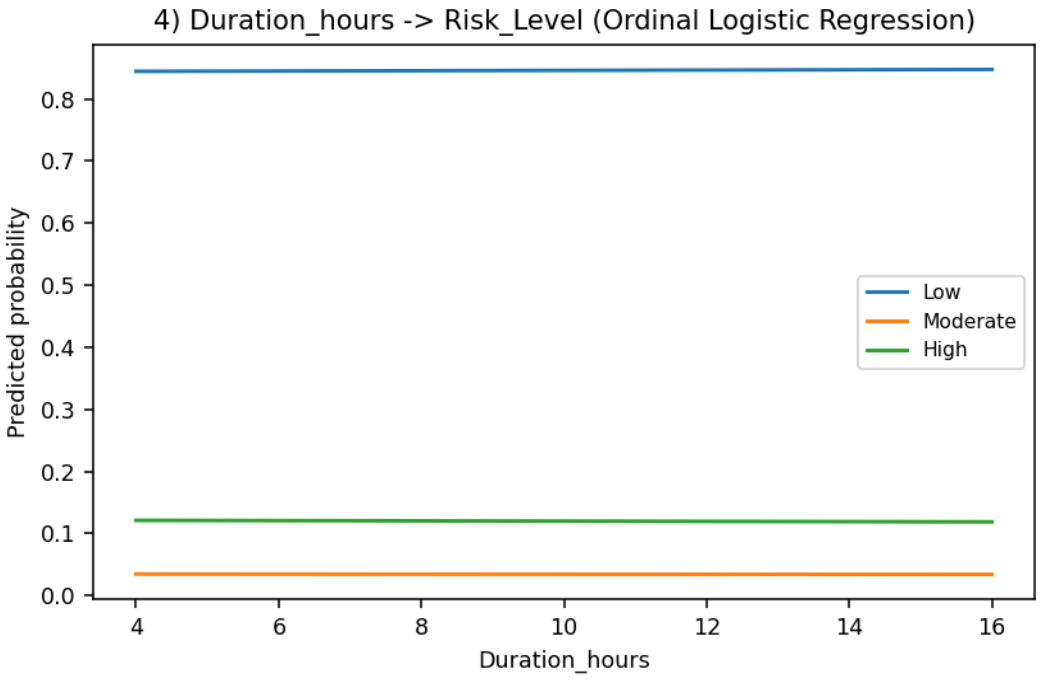
Ordinal Logistic Regression

```
X4 = df[["Duration_hours"]]
y4 = df["Risk_Level"]

model = OrderedModel(y4, X4, distr="logit")
result = model.fit(method="bfgs", disp=False)
print(result.summary())
```

Output

```
                          OrderedModel Results
==============================================================================
Dep. Variable:             Risk_Level   Log-Likelihood:                -597.32
Model:                   OrderedModel   AIC:                             1201.
Method:            Maximum Likelihood   BIC:                             1216.
Date:                Sat, 01 Aug 2026
Time:                        13:14:01
No. Observations:                1168
Df Residuals:                    1165
Df Model:                           1
==============================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Duration_hours    -0.0020      0.023     -0.086      0.932      -0.047       0.043
Low/Moderate       1.6836      0.237      7.097      0.000       1.219       2.149
Moderate/High     -1.2344      0.156     -7.916      0.000      -1.540      -0.929
==============================================================================
```



Interpretation: The coefficient on Duration_hours is -0.0020 (p = 0.932), which is not statistically significant. Sampling duration (driven by the fixed 8h/16h/24h measurement windows) has no meaningful relationship with the resulting Risk_Level classification — risk is driven by the measured noise level, not by how long the measurement ran.

5. Longitude → Duration_hours

Simple Linear Regression

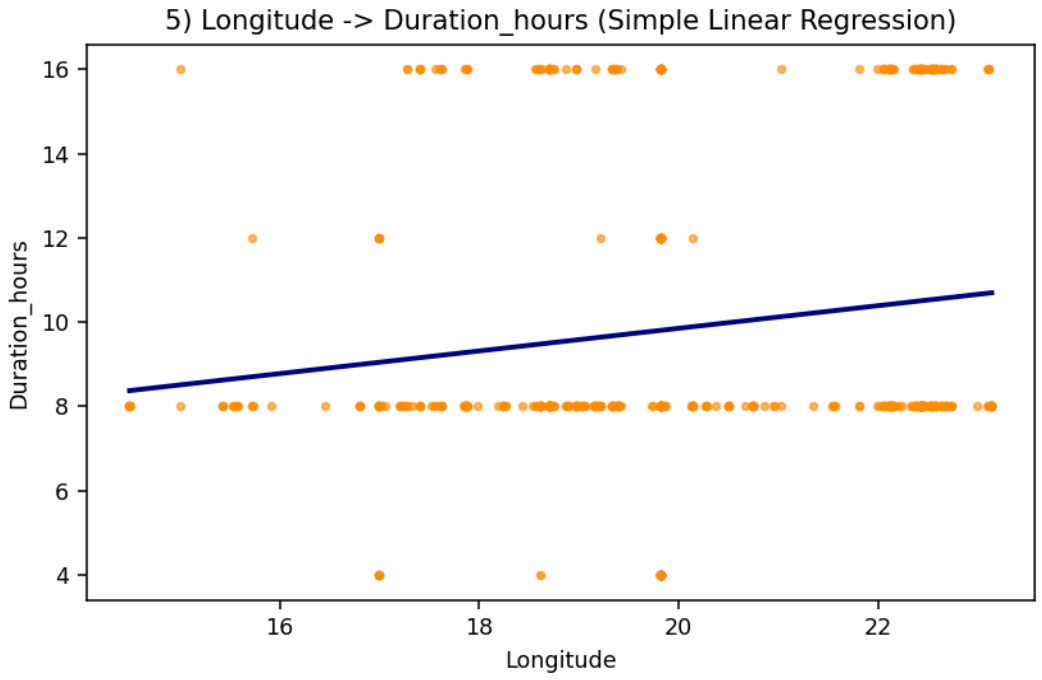
```
X5 = sm.add_constant(df["LONGITUDE"])
y5 = df["Duration_hours"]

result = sm.OLS(y5, X5).fit()
print(result.summary())
```

Output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         Duration_hours   R-squared:                       0.023
Model:                            OLS   Adj. R-squared:                  0.023
Method:                 Least Squares   F-statistic:                     28.04
Date:                Sat, 01 Aug 2026   Prob (F-statistic):           1.42e-07
Time:                        13:14:02   Log-Likelihood:                -3148.4
No. Observations:                1168   AIC:                             6301.
Df Residuals:                    1166   BIC:                             6311.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          4.4749      1.022      4.377      0.000       2.469       6.481
LONGITUDE      0.2689      0.051      5.295      0.000       0.169       0.369
==============================================================================
Omnibus:                      158.436   Durbin-Watson:                   1.906
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              150.203
Skew:                           0.807   Prob(JB):                     2.42e-33
Kurtosis:                       2.306   Cond. No.                         197.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Interpretation: Duration_hours = 4.475 + 0.2689 × LONGITUDE. R^2 = 0.023 (p = 1.42e-07) — statistically significant but weak: longitude explains only about 2.3% of the variance in measurement duration. Measuring stations further east are associated with slightly longer sampling windows, but the effect size is small.